# WARP: Optics, Holograms, and Worldlines over Shared Causal History

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**Abstract**

We define WARP as an optic architecture over shared causal history. A WARP optic places an observer-relative aperture over a bounded slice of causal history, lowers the resulting weave under an admissibility law, and retains a holographic witness sufficient for replay, audit, transport, and bounded revelation. Deterministic ticks, braid collapse, and replica import are thus instances of one recurring act: slice, lower, witness, retain.

We argue that WARP is closed over its own witness-bearing outputs. A tick receipt, braid shell, import shell, or property certificate is itself a first-class object for later folding, revelation, comparison, or reliance, not an after-the-fact log attached to an execution.

The distributed consequence is direct: network distance does not introduce a second admission law. Transport constructs the comparable slice on which replica-scale lowering can occur; once that slice exists, the downstream optic is the same one already used locally. Distribution changes the slice boundary; it does not change the downstream law.

# Contents

## 1 Introduction

The preceding AIΩN papers [Ros25a, Ros25b, Ros25c, Ros25d, Ros26a, Ros26b] established the mechanisms that make WARP distinctive: deterministic execution, provenance-complete boundary artefacts, observer-relative revelation, executable branching over shared prefixes, and the sovereignty pressure created by replayable shared history. This paper does not add another mechanism. It names the control shape that makes those mechanisms one architecture. That object is the WARP optic (Definition 3.1).

The optic is a composite act. It begins with a positioned observer and a weave of plural claims, constructs a bounded slice on which judgement is lawful, admits a witnessed outcome under policy, and retains a holographic boundary for later replay, audit, replica transport, or bounded revelation (Section 3). The same act recurs at tick, braid, and replica scale (Section 4), and we ground each scale in a worked example (Section 5).

We use a direct noun map and then enforce it. A worldline carries admitted causal history. A weave holds basis-relative claims in suspension. An optic lowers that weave through a bounded aperture under policy. A hologram retains the witness-bearing boundary of the lowering. These nouns are not synonyms for older rewrite vocabulary; they separate responsibilities that otherwise blur together in implementation: basis construction, lawful judgement, and retained obligations.

We also state one ontology choice early, because it stabilises the rest of the paper. We treat witnessed causal history as primary: admitted transitions, frontiers, witnesses, and retained boundary artefacts are the primitive objects. "The graph" then names a frontier-relative chart over that history, emitted by an observer under a lawful basis and aperture. This makes WARP graph-expressive without committing the architecture to a privileged materialised graph object underneath it.

Two consequences are derived and then reused throughout. First, WARP is closed over its own witness-bearing outputs (Definition 3.5): receipts, braid shells, import shells, and certificates are first-class objects for later folding, revelation, comparison, or certification. Secondly, distribution changes what must be normalised to become comparable before judgement; it does not introduce a second admission law (Section 4 and Example 3).

Reducer or rewrite language explains a local transformation. The optic form explains the larger control law around that transformation: bounded focus, lawful judgement, retained residual, and reuse across execution, replay, explanation, transport, and revelation. It is also the reason we adopt the profunctor form in Section 3.1: the same declared optic should support multiple interpretations without changing its structural focus.

**Position in the series and neighbouring work.** This paper extracts the invariant the earlier foundations papers approached in part. The present discussion therefore sits at the intersection of algebraic graph transformation [EEPT06, CEL+97], event structures and causal partial orders [Lam78, Win87], process semantics [Hoa85, Mil89, Par81], provenance systems [BKT01, CCT09], and distributed reconciliation [DHJ+07, SPBZ11, GL02]. Its contribution is the identification of the optic-shaped law that recurs across them and the architectural consequences of taking that law seriously.

**Contributions.** The specific contributions are:

1. An optic-centered reframing of WARP around four primary nouns: worldlines, weaves, optics, and holograms.
2. A native definition of the WARP optic, separating observer structure, bounded aperture, set-side lowering, admissibility law, and retention contract.

3. A clarification of why the optic form matters: the same declared WARP act may lawfully support execution, footprint derivation, witness extraction, debugger explanation, translation analysis, and schema compilation—that is, compilation of app-authored optic surfaces into runtime-checkable intent and observer artefacts—without changing its structural focus.
4. A scale-by-scale identification of the same optic-shaped lowering in three primary settings: local tick execution, braid-local collapse, and distributed suffix import after transport to a common basis.
5. A recursive hologram result: each such lowering emits a boundary-complete shell for the bounded admitted situation, so downstream replay and counterfactual examination propagate beyond ticks to braid-local, replica-local, and later trust-bearing shells.
6. A normalisation thesis for distributed causality: network distance relocates the principal burden into transport and normalisation to a comparable frontier.
7. A three-way architectural decomposition into commitment, folding, and revelation, separating what is true from how it is stored and who may lawfully inspect it.
8. A post-Unix reframing in which processes and files are treated as materialised views over shared causal history, together with a protected working interval spanning ephemeral, author-only, and shared lanes.
9. An architectural bridge from provenance to privacy through property certificates, reliance status, and observer rights, showing how bounded revelation can coexist with replayable history.

**Scope.** This paper is architectural, with a small amount of formal structure. Its goal is to extract the common optic shape that recurs across the WARP stack and to show how that shape supports both strong provenance and bounded revelation. The remaining frontier laws opened by this view—including contamination calculus, collective sovereignty, export semantics, and hostile transport conditions—remain follow-on work. The present paper fixes the architectural centre with sufficient precision that those later laws can be treated as extensions. Accordingly, most formal statements in this paper are propositions with argument sketches about recurrent structure; only the replica-scale confluence result is stated as a theorem because it makes a specific behavioural claim about retained shells.

**Reading guide.** Section 2 stabilises the noun map. Section 3 gives the core optic definition and the local laws needed to read it. Section 4 then states the recurrent scale-local claims. The remaining sections draw distributed, systems, and governance consequences from that control shape.

## 2 Causal Lanes, Strands, and Braids

Before defining the optic, we need the basic ontology. The earlier papers developed deterministic execution, branching, and replay, but they did not need to stabilise the noun map at this level of explicitness. Here we do, because the same optic acts over different kinds of objects, and those objects must not be allowed to collapse into one another.

**Definition 2.1** (Causal lane)**.** A *causal lane* is a replayable carrier of causal history equipped with a frontier-relative materialisation map. At minimum, a lane has an identity, an event or tick order, a causal provenance record, and a rule by which a frontier or tick determines a materialised state.

This use of lanes is intentionally close to the older language of events, causal dependence, and partial orders: it assumes that a computation admits multiple serialisations, with its causal structure captured by partial orders [Lam78, Win87].

The term *lane* is therefore the ontology-level noun for a history-bearing computational object. It is intentionally broader than *worldline*. A lane may be canonical or speculative; it may already be socially admitted, or it may still be a private counterfactual carrier.

**Definition 2.2** (Worldline)**.** A *worldline* is a canonically admitted causal lane. It is the lane form whose history and frontier-relative materialisations count as shared causal reality for the relevant policy frame.

**Definition 2.3** (Strand)**.** A *strand* is a speculative causal lane rooted at a parent worldline and a fork anchor. A strand carries fork provenance back to that parent anchor, together with whatever local history is required for the strand to diverge lawfully from the inherited base.

Two consequences should be kept explicit. First, a strand is a real lane with causal identity and fork provenance. Secondly, its definition is a divergence record that can be resolved at multiple bases. Its materialisation at a given frontier is obtained by resolving the strand's local divergence against the inherited parent history at the chosen basis.

Operationally, this means a strand is read holographically. A bounded read of a strand materialises only the backward causal cone required by the strand's local divergence and optic footprint at the chosen basis. The strand is real as a speculative lane; its realised state is a frontier-relative reading over inherited history plus local divergence, and it avoids eager copying of the whole parent lane.

**Definition 2.4** (Braid)**.** A *braid* is a lane-scale weave: a plural composition over a family of lanes sharing a common precursor or comparison basis. It presents multiple validated lane claims over a common frontier so that a later collapse or admission act may judge them.

**Definition 2.5** (Weave)**.** A *weave* is a basis-relative carrier that holds multiple candidate claims in suspension without yet forcing a single derived answer. Candidate rewrite bundles, braids, and transported import families are the three primary weaves studied in this paper.

The name is geometric: we use "braid" because lane traces literally interweave over a shared basis. Full braid-group machinery is outside the present scope [Art47]. This is also the first place where ordinary version-control language becomes misleading. A braid is not a latent merge result. It is the weave over which collapse may later derive a lane-level result, preserve plurality, surface conflict, or return obstruction. In that respect it is closer to patch-theoretic comparison objects than to a preordained merge target [MDG13].

*Remark* 2.6 (Ontology versus realisation). These nouns are ontology-level nouns. Realisation-level objects are frontier-relative views: a strand resolved at a particular parent frontier, or a braid presented over a specific common basis. This distinction is necessary because a strand may be a stable noun while its realised view changes with the frontier at which it is interpreted.

The same split also prevents admission from silently doing other jobs. Forking creates ancestry. Braiding constructs a weave. Collapse derives a lane-level result from that weave when policy and witness conditions allow it. Admission, finally, blesses a derived result as canonical when the governing law requires that further step. If these verbs are allowed to blur together, the rest of the paper becomes incoherent.

**Operational intuition.** A worldline is the shared lane others may rely on. A strand is the speculative lane in which counterfactual or author-local work may proceed without immediately becoming shared fact. A braid is the object that appears when several such lanes must be compared over one basis without yet pretending they have collapsed into a single answer. The optic introduced next acts over all three scales, but it acts over a different weave at each scale.

**Specification note.** We keep the runtime interface out of line. The noun map is implementation-bearing, but the concrete interface depends on the host language and toolchain. In the current stack, authored SDL boundaries are compiled by a schema compiler into generated host bindings and runtime-checkable footprint, witness, and retention obligations. The present paper fixes the invariant control shape that those compilations implement.

**Terminology summary.**

| Terminology | Definition | Remarks |
|---|---|---|
| Causal lane | Replayable carrier of causal history with frontier-relative materialisation. | May be canonical or speculative. |
| Worldline | Canonically admitted causal lane. | Counts as shared causal reality under the governing policy frame. |
| Strand | Speculative causal lane rooted at a parent worldline and fork anchor. | Resolves divergence against a chosen basis. |
| Braid | Lane-scale weave over multiple lanes sharing a common precursor or basis. | Comparison carrier presented for later collapse or admission. |
| Weave | Basis-relative carrier of suspended candidate claims. | Tick bundles, braids, and transported import families are the primary cases. |
| Hologram | Retained boundary sufficient to recover the optic-accurate bulk under a retention contract. | Shell $\theta$ is a concrete carrier of a hologram. |

## 3 The WARP Optic

Our main object is the WARP optic. It names the composite act from which local execution, braid collapse, replica import, replay, debugger explanation, and witness retention are all built. A WARP optic couples a positioned observer to a bounded site, a lawful set-side lowering, and the retained boundary that later makes that act replayable, auditable, transportable, or revealable.

**Definition 3.1** (WARP optic)**.** Fix an observer plan $\Omega = (\pi, \beta, M, U, E)$. A WARP *optic* is a tuple

$$\Psi = (\Omega, \chi, \rho, \Pi, \Lambda)$$

where:

- $\Omega$ fixes the revelation-side observer discipline: projection, basis, observer state, update discipline, and emission;
- $\chi$ is the bounded aperture or optic slice within which a local judgement is lawful;
- $\rho$ is the set-side lowering surface that presents a comparable claim surface over $\chi$ from a frontier-relative weave;
- $\Pi$ is the admissibility law governing what may be derived, preserved as plural, surfaced as conflict, or obstructed at that slice;
- $\Lambda$ is the retention contract specifying which replay, audit, transport, revelation, or reliance obligations the retained shell must still satisfy.

A WARP optic acts on a basis-relative weave $\mathcal{P}$. We write

$$\mathrm{Lower}_\Psi(F_*, \mathcal{P}) = (R, W, \theta)$$

with the internal factorisation

$$(C, \chi) = \rho(F_*, \mathcal{P}; \Omega), \qquad (R, W) = \mathrm{Admit}_\Pi(F_*, C, \chi), \qquad \theta = \mathrm{Pack}_\Lambda(R, W).$$

The internal names remain useful. We treat the optic as the primary object, with these helper acts serving as its internal factorisation.

Lowering names the composite optic act. Admission names the policy judgement inside that act. WARP therefore remains an admission architecture: the optic first constructs a comparable slice, then admits, preserves plurality, surfaces conflict, or obstructs under policy, and finally packs the witnessed outcome into a retained shell.

**Tiny worked instance.** At tick scale, one can read the factorisation almost literally. Let $\mathcal{P} = \{c_1, c_2\}$ be two candidate rewrites whose footprint overlaps at a node $n$. Then $\rho$ presents the local claim surface over the footprint slice $\chi = \mathrm{footprint}(n)$, admission returns a derived outcome plus a witness of the blocked coexistence, and packing retains a receipt-like shell:

$$(C, \chi) = \rho(F_*, \{c_1, c_2\}; \Omega), \quad (R, W) = \mathrm{Admit}_\Pi(F_*, C, \chi), \quad \theta = \mathrm{Pack}_\Lambda(R, W).$$

The point is not the toy $n$, but the separation of responsibilities: basis construction, lawful judgement, then retention.

This presentation also keeps the set side architectural. One runtime may realise $\rho$ by bounded DPO graph rewriting; another may realise it by CRDT-style set semantics over an append-only history. We name the invariant role and leave the specific calculus open.

**Three immediate specialisations.** The definition is intentionally generic, but it should already read concretely in the three primary cases:

- at tick scale, $\mathcal{P} = C_{\mathrm{rw}}$ is the candidate rewrite bundle and $\rho$ returns the local footprint slice over one current frontier;
- at braid scale, $\mathcal{P} = B$ is the braid and $\rho$ returns the common-basis comparison slice over the participating strands;
- at replica scale, $\mathcal{P} = \widetilde{C}$ is the transported remote suffix family and $\rho$ returns the imported-region slice at the receiver's comparable frontier.

The optic law is the same in all three cases. What changes is the weave and the work needed to present a comparable slice for lawful judgement.

**Definition 3.2** (Optic slice)**.** The bounded site $\chi$ is the *optic slice*: the least frontier-relative closure over which the governing law can lawfully compare the claims in the weave and judge their coexistence, derivation, plurality, conflict, or obstruction. The lowering surface produces claims over such a slice; the hologram later preserves the result slice needed for later replay and audit, transport, or bounded revelation. This generalises the spirit of program slicing—isolating what is relevant to a criterion—from program statements to witnessed causal history [Wei81].

We make the outcome space explicit with one typing. Let $X$ be the scale-local carrier of derived results. Then

$$\mathcal{O}(X) := \mathrm{Derived}(X) \;\sqcup\; \mathrm{Plural}(X) \;\sqcup\; \mathrm{Conflict} \;\sqcup\; \mathrm{Obstruction}\,.$$

Optic lowering returns a witnessed member of an outcome algebra.

Truth should be read operationally here. In WARP, "truth" names what may be lawfully admitted relative to a frontier, an optic, and a witness. Lowering therefore supports more than binary accept or reject: lawful outcomes may also preserve plurality, surface conflict, or return obstruction.

### 3.1 Why The Optic Form Matters

The optic form matters because the same declared optic should support more than one interpretation. In profunctor form, $\mathsf{Opt} : p\,a\,b \to p\,s\,t$, varying the profunctor $p$ changes how the act is interpreted while leaving its structural focus unchanged. For WARP, we use this to reuse one declared optic for execution, footprint and slice derivation, witness extraction or validation, debugger explanation, translation analysis, and compilation.

The term *optic* is used in deliberate continuity with the lens and profunctor-optics tradition, in which a focused transformation can be lifted into a transformation of a larger whole [FGM+07, PGW17, BG18]. WARP uses the optic discipline architecturally. We separate focus and residual from lawful judgement, witness, and observer-relative emission, so that execution, replay, explanation, and revelation can share one declared structure.

In profunctor notation, the core optic idea is:

$$\mathsf{Opt} : p\,a\,b \to p\,s\,t$$

where $s$ and $t$ are the whole before and after the act, $a$ and $b$ are the focused part before and after the act, and $p$ is an abstract transformation context. Read operationally: if one can transform the focused part from $a$ to $b$ inside some lawful context $p$, the optic lifts that into a transformation of the whole from $s$ to $t$.

For WARP, the important consequence is that the same declared optic can be interpreted multiple ways without changing its structural focus: the same act may be executed as a lowering, read as a slice or footprint derivation, used to extract or validate witness structure, rendered as a debugger explanation, analysed for conflict or translation cost, or compiled into runtime artefacts. That reuse is one of the main reasons the paper needs optic language in addition to reducer language.

### 3.2 Residual, Witness, and Hologram

The optic story includes a hidden residual. For a lens-like optic, one standard representation is [PGW17, BG18]

$$\mathrm{Lens}(S,T,A,B) \cong \int^{M} \mathcal{C}(S, M \times A) \times \mathcal{C}(M \times B, T).$$

This says that the whole $S$ can be decomposed into a focused part $A$ and a hidden residual $M$, then reassembled from the updated focus $B$ and that same residual. The coend matters because the optic semantics does not care about one privileged presentation of the residual. It quotients over equivalent presentations of the hidden context.

That perspective is exactly the right intuition for WARP shell equivalence. Two retained shells may preserve the same optic-accurate bulk of a bounded admitted situation while differing in their concrete storage shape. The witness $W$ is the residue explaining why the lowering was lawful; in the constructive tradition, it is closer to a proof-bearing object than to an after-the-fact annotation [ML84]. The hologram $\theta$ is the retained carrier. A Boundary Transition Record is therefore not the optic law itself. It is one concrete shell family carrying the law's consequences for later use.

In lens language the residual $M$ is typically hidden. In WARP, the residual is made explicit and split: $\theta$ is the retained carrier, $W$ is the witness-bearing residue explaining lawfulness, and $\Lambda$ names the future obligations that equivalence must preserve.

**Definition 3.3** (Optic-accurate bulk)**.** For optic $\Psi$ and retention contract $\Lambda$, the *optic-accurate bulk* of a bounded admitted situation is the portion of that situation, together with the witness structure it requires, that must remain recoverable for the lawful observations and obligations named by $\Psi$ and $\Lambda$. It is a bounded notion: the retained causal content sufficient for the replay, audit, transport, revelation, or reliance uses permitted at that aperture.

**Definition 3.4** (WARP hologram)**.** For optic $\Psi$ and lowering result

$$\mathrm{Lower}_{\Psi}(F_*, \mathcal{P}) = (R, W, \theta),$$

a WARP *hologram* is the retained shell $\theta$, together with any referenced basis anchors, such that $\theta$ is sufficient to recover the optic-accurate bulk of the bounded admitted situation, as determined by $\Psi$, up to the equivalence demanded by the retention contract $\Lambda$. The contract is explicit: preserve the least sufficient boundary from which the admitted outcome and its witness-bearing consequences can be reconstructed, replayed, audited, transported, or revealed as $\Lambda$ requires.

**Proposition 3.5** (Closure over witness-bearing outputs)**.** WARP is closed over its own witness-bearing outputs. For each primary scale $s \in \{\mathrm{tick}, \mathrm{braid}, \mathrm{replica}\}$, there exists a scale-local optic $\Psi_s$ and a scale-local weave $\mathcal{P}_s$ such that

$$\mathrm{Lower}_{\Psi_s}(F_s, \mathcal{P}_s) = (R_s, W_s, \theta_s),$$

and the resulting hologram $\theta_s$ is itself a first-class object of later folding, revelation, certification, or further lawful comparison.

*Argument sketch.* At tick scale, the weave is the local candidate rewrite bundle; the hologram preserves the frontier coordinate, the admitted result, and the witness explaining coexistence or blockage for one lawful step. At braid scale, the weave is the braid over a common basis; the hologram preserves the bounded comparison geometry and the lawful collapse or plural outcome. At replica scale, the weave is the remote suffix family as re-expressed at a comparable frontier; the hologram preserves the imported region, the import outcome, and the transport/import witness. In each case the retained shell is a first-class WARP object, so later lawful acts can operate over it without leaving the ontology of witnessed causal history. □

This closure is what makes the architecture self-similar across scale. A local tick receipt, a braid-collapse artefact, a transport/import shell, and a trust certificate are members of one shell family: retained holographic boundaries over bounded admitted situations. Because each recurrence emits such a hologram, deterministic replay and counterfactual examination recur with the law itself. Each retained slice stores the least sufficient history of its bounded local situation.

## 3.3 Commitment, Folding, Revelation

The optic architecture becomes easier to navigate once the stack is divided into three moments:

- **Commitment**: a weave is lowered into frontier-relative truth under a governing optic.
- **Folding**: already-admitted history is re-expressed in a boundary-equivalent retained shell.
- **Revelation**: admitted or folded history is exposed to a bounded observer under an aperture and authority regime.

This three-way split is one of the main architectural results. It keeps the system from conflating what is true, how truth is stored, and who may see it. Commitment records the witnessed outcome of lawful judgement: derivation, preserved plurality, conflict, or obstruction.

The equivalence relation is intentionally behavioural: it preserves the lawful observations and replay obligations promised by a retained contract. In that sense it is closer to bisimulation and observational equivalence than to storage equality [Par81, Mil89].

**Definition 3.6** (Shell equivalence)**.** Let $\Lambda$ be a retained contract specifying which replay, audit, transport, or revelation obligations must remain lawful. Two shells $\theta$ and $\theta'$ satisfy

$$\theta \simeq_\Lambda \theta'$$

iff they are both sufficient to recover the same optic-accurate bulk of the same admitted local situation for the lawful uses demanded by $\Lambda$.

We give one small example. A tick receipt that stores an inverse fragment inline and a folded shell that stores only digest-plus-pointer references may still be $\Lambda$-equivalent for replay and audit if both reconstruct the same local rewrite, witness anchors, and retained obligations. They cease to be equivalent once $\Lambda$ demands a lawful use—for example immediate plaintext revelation—that only one of them can satisfy.

For later use, the three moments can be written as typed acts:

$$\mathrm{Commit}_\Psi(F_*, \mathcal{P}) := (R, W, \theta)$$

where

$$(R, W, \theta) = \mathrm{Lower}_\Psi(F_*, \mathcal{P}).$$

$$\mathrm{Fold}_{\Lambda}(\theta) = \theta' \qquad \text{with} \qquad \theta' \simeq_{\Lambda} \theta.$$

$$\mathrm{Reveal}_{\Omega,\alpha}(m, \theta) = (m', y, W_{\mathrm{rev}})$$

where $\Omega = (\pi, \beta, M, U, E)$ is an observer plan, $\alpha$ is the active authority regime, $m \in M$ is the current observer state, and $y$ is the emitted reading. The observer plan is the revelation-side face of the optic: it fixes what may be projected and emitted, while commitment and holographic retention preserve the set-side history on which that reading depends.

**Proposition 3.7** (Folding preserves lawful readings)**.** If $\theta' = \mathrm{Fold}_{\Lambda}(\theta)$ and the retained contract $\Lambda$ is sufficient for observer plan $\Omega$ under authority regime $\alpha$, then

$$\mathrm{Reveal}_{\Omega,\alpha}(m, \theta') \approx_{\Omega,\alpha} \mathrm{Reveal}_{\Omega,\alpha}(m, \theta)$$

where $\approx_{\Omega,\alpha}$ identifies readings that emit the same lawful observation at that aperture, basis, and authority tier.

*Argument sketch.* By shell equivalence, $\theta$ and $\theta'$ preserve the same admitted local situation for the uses demanded by $\Lambda$. If $\Lambda$ covers the observer/authority pair $(\Omega, \alpha)$, then revelation over either shell has the same lawful reading consequences up to the observer's own reading equivalence. $\square$

In short: commitment changes what may count as shared truth, folding changes the carrier while preserving an explicit contract, and revelation changes what an observer may lawfully emit from that carrier without changing the admitted situation itself.

**Minimal optic form.** The runtime-facing shape is better stated language-neutrally:

$$\mathcal{P} \xrightarrow{\text{project / slice / normalise}} (C, \chi) \xrightarrow{\text{lawful lowering}} (R, W) \xrightarrow{\text{retain}} \theta.$$

The key point is structural: the runtime keeps the raw weave, comparable slice, admitted outcome, semantic witness, and retained shell as distinct objects instead of collapsing them into one undifferentiated record.

## 4 Tick, Braid, and Replica

This section turns from the core definition to the recurrent formal claims. The propositions in this section state the control shape shared across scale; the later sections derive distributed, systems, and governance consequences from that shape.

The optic recurs most cleanly in three primary instantiations.

**Proposition 4.1** (Optic-shape invariance)**.** For each primary scale $s \in \{\text{tick}, \text{braid}, \text{replica}\}$, there exist a scale-local optic $\Psi_s$, a scale-local frontier $F_s$, a scale-local weave $\mathcal{P}_s$, and a scale-local carrier $X_s$ such that

$$\text{Lower}_{\Psi_s}(F_s, \mathcal{P}_s) = (R_s, W_s, \theta_s), \qquad R_s \in \mathcal{O}(X_s).$$

*Argument sketch.* At tick scale, the weave is a candidate rewrite bundle over one current frontier. At braid scale, it is a braid presenting multiple valid strand claims over one common basis. At replica scale, it is a remote suffix family presented for import at a receiving frontier. In each case the same optical shape applies: a positioned observer and bounded aperture, a set-side lowering surface producing comparable claims, a lawful judgement under policy, and a retained hologram for later use. □

**Corollary 4.2** (Basis construction and slicing)**.** For each primary scale, the set-side lowering surface $\rho_s$ begins by constructing a comparable bounded slice. At tick scale, $\rho_{\text{tick}}$ is the identity on already local candidate rewrites. At braid scale, $\rho_{\text{braid}}$ presents validated strand claims over a common basis. At replica scale, $\rho_{\text{replica}}$ transports remote suffixes to a comparable frontier before any import judgement is attempted.

*Argument sketch.* The optic never judges the whole world at once. It only judges the bounded, comparable closure required for the act at hand. At replica scale that makes transport the principal burden, because the comparable slice must first be constructed. At tick and braid scale the relevant slice is already closer to hand. □

This is the operational sense in which WARP is a slicing architecture. At tick scale, $\chi$ is the footprint closure: the local neighbourhood needed to judge coexistence. At braid scale, $\chi$ is the refined comparison slice over a common lane basis. At replica scale, $\chi$ is the imported-region slice produced after transport has re-expressed the remote strand fragment at a comparable frontier. The optic therefore takes slices in and emits slices out: the set-side surface constructs the bounded site on which judgement is lawful, and the hologram retains the least sufficient boundary from which that judged situation can later be replayed, audited, transported, or revealed.

The practical consequence is that distance changes the basis construction required before the optic can lower the weave. Distribution is often treated as a distinct reconciliation regime when comparability, basis, and witness obligations remain implicit. WARP promotes those preconditions into explicit structure and then reuses the same witnessed optic. Distribution shifts the slice boundary while leaving the downstream law unchanged. The principal normalisation burden therefore receives a precise address: transport and $\rho_{\text{replica}}$ absorb the work of bringing remote claims to a comparable frontier.

| Scale | Weave | Slice | Basis construction | Shell |
|---|---|---|---|---|
| Tick | rewrite bundle | footprint closure | already local projection | tick receipt |
| Braid | strand braid | comparison cells | common-basis refinement | braid shell |
| Replica | remote suffix family | imported region | transport to comparable frontier | import shell |

All three return an element of $\mathcal{O}(X) = \text{Derived}(X) \sqcup \text{Plural}(X) \sqcup \text{Conflict} \sqcup \text{Obstruction}.$

### 4.1 The Local Tick Optic

At engine scale, the frontier is the current state, the weave object is the candidate rewrite bundle, the optic slice is the local footprint closure, and the governing policy is the scheduler or local coexistence law for that tick. The set-side role is architectural; one runtime may realise it by bounded DPO graph rewriting, while another may realise it by CRDT-style set semantics. What matters here is the invariant optic contract.

$$\mathrm{Lower}_{\Psi_{\mathrm{tick}}}(F_{\mathrm{tick}}, C_{\mathrm{rw}}) = (R_{\mathrm{tick}}, W_{\mathrm{tick}}, \theta_{\mathrm{tick}}), \qquad R_{\mathrm{tick}} \in \mathcal{O}(X_{\mathrm{rw}}).$$

The tick-local hologram $\theta_{\mathrm{tick}}$ preserves enough boundary information to replay the lawful local step and its blocked alternatives without materialising the whole host history.

### 4.2 The Braid Optic

At lane scale, the frontier is a common parent frontier, the weave object is a braid, the optic slice is the refined braid-cell family, and the governing policy is the braid-scale collapse law. A braid is the structured weave over which the optic lowers. Its outcome algebra remains `Derived`, `Plural`, `Conflict`, or `Obstruction`.

$$\mathrm{Lower}_{\Psi_{\mathrm{braid}}}(F_{\mathrm{braid}}, B) = (R_{\mathrm{braid}}, W_{\mathrm{braid}}, \theta_{\mathrm{braid}}), \qquad R_{\mathrm{braid}} \in \mathcal{O}(X_{\mathrm{lane}}).$$

The braid-level hologram $\theta_{\mathrm{braid}}$ preserves the comparison basis, the plural claim geometry relevant to the bounded cells, the admitted collapse outcome, and the witness that makes that outcome reviewable or transportable without forcing premature flattening.

This sharpens the paper's branching vocabulary. Irreducible plurality need not be treated as merge failure; under the governing law it may itself be the lawful lowered result. That contrasts with convergence-first replicated editing and CRDT presentations whose central obligation is to transform concurrent operations into one consistent visible state [EG89, SPBZ11].

### 4.3 The Replica Optic

At replica scale, the frontier is the common replicated prefix, the weave is the remote suffix family, the optic slice is the affected imported region, and the governing policy is the local transport-and-import law. This sits downstream of Lamport's causal ordering insight and of the later replicated-state lineage, but relocates the hard work into explicit transport and witnessed import [Lam78, DHJ+07, SPBZ11]. The set-side surface $\rho_{\mathrm{replica}}$ constructs the comparable basis on which the replica-scale weave exists:

**Frontier legend.** We write $F_{\mathrm{rep}}$ for a shared replicated prefix (the latest frontier both sides can still name as common ancestry), and $F_{\mathrm{recv}}$ for the receiver's current comparable frontier at which the transported suffix is to be judged. Transport re-expresses remote claims at $F_{\mathrm{recv}}$; lowering then occurs at that receiving frontier, not at the shared prefix.

$$\widetilde{C}_{\mathrm{rep}} := \mathrm{Transport}(F_{\mathrm{rep}}, \Sigma_{\mathrm{remote}}, F_{\mathrm{recv}}).$$

Only after that basis construction can the optic be lowered:

$$\mathrm{Lower}_{\Psi_{\mathrm{rep}}}(F_{\mathrm{recv}}, \widetilde{C}_{\mathrm{rep}}) = (R_{\mathrm{rep}}, W_{\mathrm{rep}}, \theta_{\mathrm{rep}}), \qquad R_{\mathrm{rep}} \in \mathcal{O}(X_{\mathrm{imp}}).$$

The transport/import hologram $\theta_{\mathrm{rep}}$ preserves the comparable basis established by transport, the import judgement, and the witness needed for replayable catch-up, audited settlement, or bounded revelation at the receiving runtime. Transport constructs the basis on which the replica-scale optic can lower the imported situation using the same downstream law already trusted locally.

The same distributed case can be read more geometrically as strand braiding under lag. Each participant advances along a local strand whose frontier may be ahead of what the other participant can yet lawfully treat as shared. For the bounded act at hand, the common basis is therefore a lagging frontier: the latest frontier both sides can still treat as mutually comparable. An incoming network message carries a transported strand fragment from that lagging basis to the sender's newer frontier. The receiver transports that fragment to the comparable basis, braids it with its own locally advanced strand over the bounded site, and lowers that braid under the governing optic. If the lowering is admitted, the mutually relied-upon basis can advance.

**Latency and settlement.** This distinction becomes especially important once imported first-order history is separated from imported settlement over already-judged history. A runtime may lawfully compile a transported first-order intent into the same downstream tick path it uses locally, provided transport has already reconstructed the authored basis and attached the required provenance and witness conditions. But an outcome-bearing import can be treated as settlement over already-judged history. In a networked racing simulation, for example, a finish-line crossing authored at time $T_s$ may be received only at $T_s + L$. The crossing must be judged at basis $T_s$. Receive time only determines when that judgement becomes mutually visible. Once lowered, the resulting imported shell asserts a settlement fact: the earlier crossing at $T_s$ is now mutually relied upon. Such a settlement shell may update standings, UI state, or bounded present repair; it should enter the settlement path with idempotent semantics over already-adjudicated historical facts.

The theorem in this section isolates one safe confluence zone. Hostile or failure-path transport remains hard. We do not claim here that transport is trivial or fully formalised under hostile distributed conditions; Byzantine behaviour, partial connectivity, replay attacks, and other failure-path complications remain part of the normalisation frontier rather than being dissolved by the optic. In such cases, the intended outcome is not silent inconsistency but a lawful *obstruction* with a retained witness explaining why import could not be admitted (Appendix A). The statement localises that difficulty in Transport and in the replica optic's set-side lowering surface, then states the strongest behavioural claim the retained shells can support once those preconditions are satisfied.

**Theorem 4.3** (Derived independent suffix confluence up to shell equivalence)**.** Let $\widehat{C}_1$ and $\widehat{C}_2$ be two remote suffix claim families authored at source frontiers $F_1^{\mathrm{src}}$ and $F_2^{\mathrm{src}}$. Let $F^{\mathrm{recv}}$ be a receiving comparable frontier. Let

$$\widetilde{C}_i := \mathrm{Transport}(F_i^{\mathrm{src}}, \widehat{C}_i, F^{\mathrm{recv}})$$

be their transports to $F^{\mathrm{recv}}$. Let $\chi_1$ and $\chi_2$ be the imported sites of $\widetilde{C}_1$ and $\widetilde{C}_2$, and let $\chi_{12}$ be their common refinement. Assume:

1. there exist individual commitments

   $$(R_i, W_i, \theta_i) := \mathrm{Commit}_{\Psi_{\mathrm{rep}}}(F^{\mathrm{recv}}, \widetilde{C}_i)$$

   over sites $\chi_i$, with derived outcomes $R_i \in \mathrm{Derived}(X_{\mathrm{imp}})$, and let $F_i$ be the derived frontier exposed by $\theta_i$;

2. *slice independence*: the imported sites $\chi_1$ and $\chi_2$ are either disjoint or satisfy the replica overlap law required for lawful joint lowering over $\chi_{12}$;

3. *witness monotonicity*: admitting either transported family preserves, up to shell equivalence, the basis and witness conditions required to admit the other;

4. *retention stability*: composing retained shells along either sequential path preserves the retained replay and audit obligations for the imported region;

5. *transport stability*: re-normalising one raw suffix against the frontier produced by admitting the other yields a lawful follow-on claim family equivalent on the refined site to the corresponding joint transport.

Define the re-normalised follow-on families

$$\widetilde{C}_2^{(1)} := \mathrm{Transport}(F_2^{\mathrm{src}}, \widehat{C}_2, F_1), \qquad \widetilde{C}_1^{(2)} := \mathrm{Transport}(F_1^{\mathrm{src}}, \widehat{C}_1, F_2).$$

Write

$$(R_{12}, W_{12}, \theta_{12}) := \mathrm{Commit}_{\Psi_{\mathrm{rep}}}(F^{\mathrm{recv}}, \widetilde{C}_1 \sqcup \widetilde{C}_2)$$

for one-shot combined import. Write

$$(R_{1;2}, W_{1;2}, \vartheta_{1;2}) := \mathrm{Commit}_{\Psi_{\mathrm{rep}}}(F_1, \widetilde{C}_2^{(1)})$$

and

$$(R_{2;1}, W_{2;1}, \vartheta_{2;1}) := \mathrm{Commit}_{\Psi_{\mathrm{rep}}}(F_2, \widetilde{C}_1^{(2)}).$$

Let $\Theta_{1;2}$ and $\Theta_{2;1}$ denote the $\Lambda$-retained shells for the two-step paths $(\theta_1, \vartheta_{1;2})$ and $(\theta_2, \vartheta_{2;1})$, respectively. Then for any retained contract $\Lambda$ covering replay and audit of the imported region,

$$\theta_{12} \simeq_\Lambda \Theta_{1;2} \simeq_\Lambda \Theta_{2;1}.$$

*Proof sketch.* The individual commitments provide the two frontier advances $F_1$ and $F_2$. Slice independence supplies a lawful common refinement $\chi_{12}$ on which the combined import may be judged. Witness monotonicity ensures that transporting the second suffix against either derived frontier does not destroy the basis or witness conditions required for the follow-on import. Transport stability identifies the two re-normalised follow-on claim families with the jointly transported one on the refined site. Retention stability then ensures that the sequentially composed shells preserve the same replay and audit obligations as the one-shot shell under $\Lambda$. The one-shot import shell $\theta_{12}$ and the two sequential path shells $\Theta_{1;2}$ and $\Theta_{2;1}$ therefore preserve the same admitted local situation and the same retained obligations for the imported region. Hence they are shell-equivalent under $\Lambda$, even though the runtime may have reached them by different admissible paths. □

**Corollary 4.4** (Non-independent suffixes)**.** If the transported suffix families are not independent, we do not assert commutativity or confluence. The receiving runtime still evaluates the bounded imported site under the same optic law and returns a witnessed member of the outcome algebra $\mathcal{O}(X_{\mathrm{imp}})$: derived result, preserved plurality, conflict, or obstruction.

This also identifies the safe availability zone more precisely than ordinary replication slogans do. The easy distributed case is the footprint-independent fragment of the workload: imported claim families whose bounded sites can be transported and admitted without destroying one another's witness conditions.

That is why the theorem targets shell equivalence in place of naive final state agreement. Two import paths may agree on a terminal frontier while still differing in retained witness, replay obligations, or intent preservation. The architecturally relevant invariant is therefore stronger than state equality alone.

One immediate consequence is that replication state becomes a queryable temporal causal structure once remote suffixes are represented as witnessed transport objects. One should therefore expect a CTL*-like family of queries over lawful transport and lowering paths: what imports were possible, what conflicts were inevitable, what acknowledgements must arise under policy, and what an observer at a bounded aperture was entitled to see or rely on along a branch [EH86]. We do not formalise that logic here; the only point is that such queries depend on representing replication as witnessed causal structure.

## 5 Illustrative Examples

The preceding three optic instantiations are easier to retain when each case is grounded in one concrete example. We therefore record three minimal worked examples, one for each primary scale.

### Example 1: Local Overlap In One Tick

Let the current frontier $F_*$ materialise a local reading containing a node $n$ with status `draft`. Suppose the candidate bundle $C = \{c_1, c_2\}$ contains two local rewrites: $c_1$ changes $n$ to `published`, while $c_2$ changes the same node to `archived`. The bounded site $\chi$ is the semantic footprint containing $n$ together with whatever neighbourhood is required for the scheduler to judge the update lawfully. The governing policy $\Pi$ requires exclusive mutation over that footprint within a single tick.

Figure 1 shows the overlapping-site geometry.

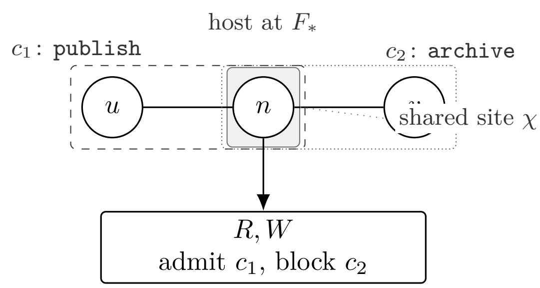


Figure 1: DPO-style overlapping matches over a shared host site [EEPT06, CEL+97]. The two candidates overlap on $n$, so admission preserves the admitted rewrite together with a witness of the blocked alternative.

In this case admission cannot lawfully admit both candidates. A typical result is that one candidate is admitted, the other is blocked, and the witness $W$ records that the candidates overlapped on the same semantic site under an exclusivity policy. **Engineering implication.** The receipt is not merely a log of what happened, but a replayable explanation of why the blocked candidate could not lawfully coexist in that tick.

In optic notation, the local scheduler lowers the candidate rewrite bundle over the shared slice $\chi$:

$$\mathrm{Lower}_{\Psi_{\mathrm{tick}}}(F_*, \{c_1, c_2\}) = (R, W, \theta), \qquad R \in \{\mathrm{Derived}(c_1), \mathrm{Derived}(c_2)\}.$$

The display says exactly that: one candidate becomes the derived result in $R$, while the overlap and blocked coexistence story are preserved in $W$ and retained in $\theta$, avoiding silent discard.

### Example 2: A Braid That Remains Plural

Let two strands $S_1$ and $S_2$ share a common parent frontier $F_*$. Each strand carries a valid claim over the same inherited object, but the claims are incompatible: $S_1$ records one lawful update to the object, while $S_2$ records another lawful update that cannot be jointly derived under the braid collapse policy $\Pi$. The refined braid cells $\chi$ isolate the shared site together with the local reintegration seams needed for comparison.

Figure 2 shows the braid geometry and the preserved plural outcome.

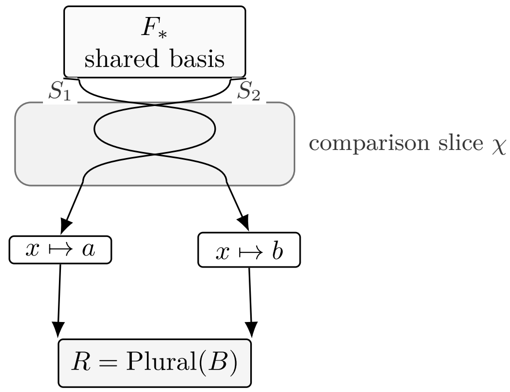


Figure 2: Two strands literally braid over a shared basis. The resulting claims remain plural when no lawful collapse exists under the braid policy.

The important point is that the lawful outcome may remain `Plural`. When collapse lacks lawful justification to choose between the two claims, plurality itself is the correct admitted result under the governing policy. The witness then preserves both the fact of divergence and the fact that preserved plurality was the lawful outcome. **Engineering implication.** Branch structure can remain representationally honest: disagreement can persist as plurality instead of being forced into a premature merge.

As a lowering display, collapse over the braid looks like this:

$$\text{Lower}_{\Psi_{\text{braid}}}(F_*, B) = (R, W, \theta), \qquad R = \text{Plural}(B).$$

The display shows that $R$ may remain plural. Plurality is itself a lawful lowered result, with the witness and retained hologram explaining why further collapse lacked justification over that bounded comparison slice.

### Example 3: Transport Before Import

Let replicas $R_A$ and $R_B$ share a common prefix $F_*$. Replica $R_A$ has admitted a suffix $\sigma_A$ beyond that prefix, and replica $R_B$ wishes to import it. Let $F_B$ denote $R_B$'s present comparable frontier. The relevant weave is the family of transported remote claims derived from $\sigma_A$. The bounded site $\chi$ is the imported region together with the local context needed to decide whether the import remains lawful at $R_B$'s present basis.

Figure 3 makes the transport-before-import order explicit.

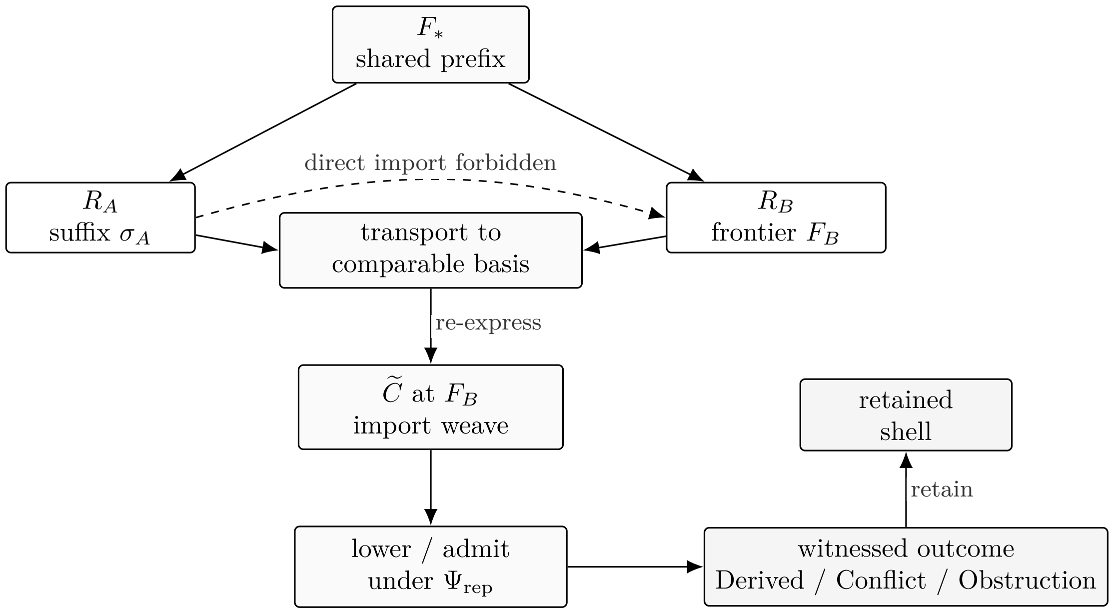


Figure 3: A remote suffix cannot be judged directly at the receiver's tip; it must first be transported to a comparable basis.

Transport first re-expresses the remote suffix at a comparable frontier. Once that transported weave exists, the replica optic can lawfully lower it. If transport or lowering fails, the result is conflict or obstruction with an explicit witness. **Engineering implication.** Replication becomes queryable and auditable as lawful transport and lowering.

At replica scale, the order of acts matters:

$$
\begin{aligned}
\widetilde{C} &= \mathrm{Transport}(F_*, \sigma_A, F_B), \\
(R, W, \theta) &= \mathrm{Lower}_{\Psi_{\text{rep}}}(F_B, \widetilde{C}), \\
R &\in \{\mathrm{Derived}(x), \mathrm{Conflict}, \mathrm{Obstruction}\}.
\end{aligned}
$$

The first line localises the principal burden in transport: the remote suffix is re-expressed at a comparable frontier before any import judgement is attempted. The second line then lowers the replica-scale weave under the same optic shape already trusted locally. The third line makes explicit that even after successful normalisation, the lowered outcome may still be derivation, conflict, or obstruction.

## 6 Continuum and the Post-Unix Horizon

We use *Continuum* to name the protocol-shaped causal medium induced by witnessed admission. It is formed by exchangeable witness-bearing artefacts—suffixes, shells, receipts, and certificates—together with the transport rules that re-express remote history at comparable bases so that the downstream optic can be lowered lawfully.

Because tick admission, braid collapse, and replica import share one optic law, it becomes natural to treat operating-system objects as frontier-relative materialisations over that same causal medium.

The post-Unix claim therefore concerns frontier-relativity at the operating-system boundary rather than a graph-shaped operating system.

The relevant lineage is Unix's process/file discipline, capability systems' authority discipline, and Plan 9's namespace discipline; WARP shifts the centre of gravity from container, descriptor, or address space to witnessed causal admission over shared history [RT74, DVH66, Lev84, PPD+95].

A tempting but inadequate shorthand would be to say that a process *is* a divergent worldline. That collapses history, live realisation, and working-set materialisation into one object. The sharper formulation is that a process is a strand whose live realisation appears as a shadow working set over shared machine history. The process is the active frontier-relative manifestation of one strand over that history.

This changes the meaning of ownership, isolation, deletion, debugging, and restart. Safety is no longer a byproduct of process opacity. It becomes a property of lawful admission over semantically closed sites. If shared history is primary, the privacy problem becomes explicit as well: the system must provide a protected interval in which exploratory work does not immediately become collaborative fact.

### 6.1 Application Boundary

The same post-Unix shift also changes how applications should meet the runtime. The relevant question is no longer how to add handwritten app methods to a host. It is how to compile app-authored optic surfaces into lawful set-side intents and lawful get-side observer plans over one generic holographic runtime. In that arrangement, the application authors its own nouns, result families, and reading families; a compiler lowers those surfaces into intent envelopes, observer plans, and shared codecs; and the runtime remains generic, admitting intents, emitting outcome/receipt/hologram boundaries, and later serving bounded readings through hosted observers. We do not formalise that compiler boundary here. The architecture developed here places it at a clear engineering seam.

Figure 4 sketches that seam.

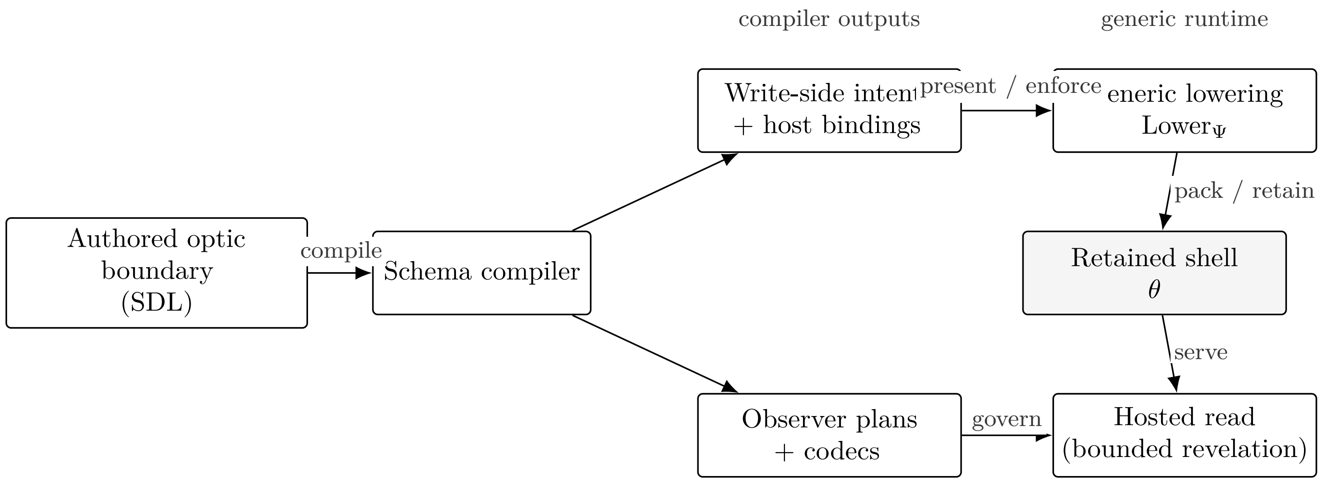


Figure 4: Authored optic boundaries compile into two runtime-facing artefacts. Write-side intents drive generic lowering. Read-side observer plans govern bounded revelation over the retained shell.

| Requirement | Prevents | Why the seam needs it |
|---|---|---|
| `REQ-SDL-1` | | |
| deterministic authored boundary | Ambient non-determinism | The runtime cannot replay, audit, or compare lowerings if authored surfaces may depend on wall-clock reads, random sources, or hidden side effects. |
| `REQ-SDL-2` | | |
| bounded footprint | Unbounded reachability | The seam must expose which nouns and closures are in play so the slice $\chi$ is finite, checkable, and lawfully lowerable. |
| `REQ-COMP-1` | | |
| checked compilation boundary | Unchecked surface-to-runtime drift | The compiler must verify slot bindings, closure obligations, and forbidden reachability before arbitrary host code runs. |
| `REQ-RT-1` | | |
| runtime gate | Bypassing admission | Runtime enforcement is required so host code cannot mutate state outside the declared optic surface. |
| `REQ-RT-2` | | |
| witness-bearing retention | Missing receipt or shell evidence | Outcomes must produce retained evidence for replay, audit, transport, and later revelation. |
| `REQ-READ-1` | | |
| rights-governed revelation | Unbounded reveal | Observer plans and authority checks must bound what a hosted read may lawfully emit from a retained shell. |

*Compiler seam requirements at the authored-boundary/runtime interface.*

### 6.2 Example: An Authored Hot-Text Optic

We make that application boundary concrete by looking at a small developer-authored hot-text boundary. The full SDL declares the hot nouns of one text layer—`BufferWorldline`, `RopeHead`, `RopeBranch`, `RopeLeaf`, `TextBlob`, `Anchor`, `Tick`, `TickReceipt`, and `Checkpoint`—together with a small family of read and write surfaces over them. The paper prints one representative rewrite and omits the full schema. Although this boundary is written in GraphQL SDL, it does not posit a canonical graph object beneath the editor; it serves instead as a convenient authoring syntax for an optic boundary: a typed aperture over one hot layer and its admissible rewrite surfaces.

GraphQL was selected here for deliberate reasons. It gives application authors one versionable schema in which to declare the typed nouns of a layer—its node/edge forms and attachments—together with the admissible rewrite surfaces over them as mutations. That in turn allows a schema compiler—in the current toolchain—to check footprint obligations, slot bindings, and forbidden reachability at compile time, before arbitrary host code has run. That check is not merely ergonomic: it rejects dishonest footprint widening before execution, because a rewrite surface cannot silently smuggle in undeclared reads, writes, creates, or closures while still claiming a narrower optic slice. Just as importantly, the SDL narrows the authoring surface: the developer is declaring shapes, closures, and mutations, rather than introducing ambient non-determinism through random sources, wall-clock reads, or hidden side effects. Because the host bindings are generated from the authored surface, they can also be checked for integrity against that surface as compiled artefacts and again when the runtime presents or enforces them. GraphQL is also readily versioned, so new nouns, rewrites, and migrations can be staged as explicit schema evolution, avoiding silent runtime drift. We treat GraphQL here as an authoring surface; its use as an API and query language is standard [Gra21].

The central rewrite in that family is `replaceRangeAsTick`, shown in Listing 1, because it shows the optic shape with minimal extra machinery:

**From surface to runtime admission.** Compilation of a surface like `replaceRangeAsTick` yields a typed intent envelope (carrying arguments plus declared reads/writes/creates/forbids) and generated host bindings that enforce those declarations and can be checked against the authored surface for integrity. At runtime, the generic lowering kernel uses the declared closures to construct the bounded slice $\chi$, applies the admissibility law $\Pi$ over the presented claim family, and packs the declared receipt and retention obligations under $\Lambda$.

```
replaceRangeAsTick(
  input: ReplaceRangeAsTickInput!
): ReplaceRangeAsTickResult!
    @wes_op(name: "replaceRangeAsTick")
    @wes_footprint(
    reads: [
      "BufferWorldline"
      "RopeHead"
      "RopeBranch"
      "RopeLeaf"
      "TextBlob"
      "Anchor"
    ]
    writes: ["BufferWorldline"]
    creates: [
      "TextBlob"
      "RopeLeaf"
      "RopeBranch"
      "RopeHead"
      "Tick"
      "TickReceipt"
    ]
    slots: [
```

```
      {
        slot: "worldline"
        kind: "BufferWorldline"
        bindFromArg: "input.worldlineId"
        access: [READ, WRITE]
      }
      {
        slot: "baseHead"
        kind: "RopeHead"
        bindFromArg: "input.baseHeadId"
        access: [READ]
      }
    ]
    closures: [
      {
        slot: "touchedRope"
        fromSlot: "baseHead"
        operator: "ropeRangeClosure"
        argBindings: ["input.startByte", "input.endByte"]
        reads: ["RopeBranch", "RopeLeaf", "TextBlob"]
        cardinality: MANY
      }
      {
        slot: "affectedAnchors"
        fromSlot: "worldline"
        operator: "anchorsIntersectingEditWindow"
        argBindings: ["baseHead", "input.startByte", "input.endByte"]
        reads: ["Anchor"]
        cardinality: MANY
      }
    ]
    createSlots: [
      { slot: "newBlob", kind: "TextBlob", cardinality: OPTIONAL }
      { slot: "newLeaves", kind: "RopeLeaf", cardinality: MANY }
      { slot: "newBranches", kind: "RopeBranch", cardinality: MANY }
      { slot: "nextHead", kind: "RopeHead" }
      { slot: "tick", kind: "Tick" }
      { slot: "receipt", kind: "TickReceipt" }
    ]
    updates: [{ slot: "worldline", fields: ["canonicalHead"] }]
    forbids: ["AstState", "Diagnostics", "GitWitness", "UiState"]
    )
```

Listing 1: An authored hot-text optic surface (`replaceRangeAsTick`) with an explicit footprint, slice closures, and forbidden reachability.

This is exactly the developer-facing form of a WARP optic. The authored surface names the hot nouns that matter, the closure operators that construct the optic slice, the created causal consequences that the runtime must retain, and the materialised reading that may change. The operation `replaceRangeAsTick` defines a bounded rewrite surface over a buffer worldline. The closures `ropeRangeClosure` and `anchorsIntersectingEditWindow` construct the local slice to be judged; the created `Tick` and `TickReceipt` make the witness-bearing consequences explicit; and the `forbids` clause makes the aperture explicit by ruling out silent dependence on AST state, diagnostics, Git witness history, or UI state.

We omit the full SDL because this excerpt already exposes the authored pattern: the developer declares the nouns in scope, the closures that build the local slice, the causal consequences that must be retained, the materialised reading that may change, and the regions that remain outside the aperture. In ordinary software a developer might declare methods over mutable structures. In a WARP-native system, the more faithful statement is that the developer authors optic boundaries and lets a schema compiler compile those boundaries into runtime-checkable footprint, witness, and retention obligations.

### 6.3 The Three-Tier Thinking Room

Continuum remains socially viable only if retention and revelation are decoupled. Shared history creates exposure pressure that a simple public/private split cannot absorb: users need room to think, to speculate, and only later to promote. We therefore distinguish a protected working interval with three practical tiers:

1. **Ephemeral Scratch**: local, weakly retained, and disposable.
2. **Author-Only Speculative Lane**: durable, replayable, sealed to the author by default.
3. **Shared / Admitted Lane**: collaboratively admitted causal truth.

This is the privacy-preserving answer to the exposure pressure created by shared causal history. The system may remember drafts, with revelation and retention governed by explicit policy and tier. Architecturally, Ephemeral Scratch is pre-admission or minimally retained local work; Author-Only lanes are durable and replayable prior to social transport; Shared lanes mark the moment private speculation crosses into collaborative truth.

This distinction matters immediately for debugger-driven counterfactual work. A reading, replay, or inspection act is revelation only. It does not by itself create new causal truth. If an operator explicitly asks to continue from an earlier coordinate or explore a counterfactual alternative, the system should emit an explicit fork rooted at an exact fork basis. Such debugger-created strands are real causal objects; they should default to Ephemeral Scratch or an Author-Only Speculative Lane until an explicit promotion moves them into shared history. When retained, their provenance should record the creating principal, tool or session origin, fork basis, and retention or revelation posture.

## 7 Privacy, Provenance, and Revelation

Privacy and provenance are not mutually exclusive [Ros26b]. We do not re-derive that machinery here. The present point is to show how its privacy, revelation, and observer-rights discipline inhabits the generic hologram and retained-contract structure of the WARP optic. Three components matter.

### 7.1 Property Library

Private speculative lanes can collaborate without exposing their full internal history. This is where proof-bearing evidence, proof-carrying certification, information-flow control, and zero-knowledge style selective disclosure touch the architecture [ML84, Nec97, Den76, GMR89]. They emit a certificate of lawfulness: property-bearing evidence that type/resource safety, admission integrity, provenance linkage, and policy conformance hold at the required tier. Such a certificate should minimally identify the shared basis to which it refers, the property set it attests, the witness or proof tier under which those properties were established, and the shell anchors needed for later replay, audit, or revocation. In that sense the certificate is a special case of the generic package $\theta$: a carrier of witness-bearing consequences upon which others may lawfully rely. Some properties may survive fold-to-opaque as shell anchors or zero-knowledge proofs; others, such as full forensic traces, require later unsealing.

One compact lifecycle makes the point concrete. A private lane may first *issue* a certificate stating, for example, that a bounded build or data preparation step satisfies a declared policy at basis $F_*$. The certificate then becomes *active* once the relevant principal, witness tier, and retention contract are all in force. A collaborator may subsequently *rely* on that certificate without receiving the sealed internal trace itself, because the relied-upon shell anchors and property set are already named. If later evidence shows that the attested condition no longer holds, the certificate may be *revoked* by a new witnessed status transition that changes future lawful reliance without deleting past history. Finally, the certificate may be *archived*, preserving its basis, attested properties, and revocation history for later audit even when it is no longer active for ordinary use.

### 7.2 Reliance Status

Once certificates exist, trust itself becomes a first-class object with a lifecycle, echoing the pressure visible in PKI and certificate revocation systems while generalising the object being trusted [CSF+08]. A certificate can be issued, activated, superseded, suspended, revoked, expired, and archived. Revocation is a new witnessed status transition about future lawful reliance, not deletion. This creates a true reliance domain within WARP and makes trust an internal witnessed object.

One concrete case is a collaborator relying on an active certificate that attests a sealed build, data preparation step, or model export at basis $F_*$. While the certificate remains active, downstream admissions may rely on that shell and its property set without reopening the sealed internal trace. If the certificate is later suspended or revoked because an upstream witness is invalidated, future lawful reliance must stop immediately even though the earlier basis, reliance events, and revocation act remain replayable and auditable.

### 7.3 Observer Rights

The subject-side counterpart is an observer rights lattice governing who may discover, inspect, rely on, unseal, replay, export, delegate, or revoke access over proof-bearing artefacts. The right local noun here is not “lens” but “reading”: an observer, equipped with a bounded reading frame, obtains a lawful projection by decoding only the least required slice of a subject’s backward causal cone. In this ontology, that reading is a coordinate chart over causal history. Different observers may therefore lawfully emit different charts over the same history, and revelation law must govern which transitions between such charts are permitted, what witness must travel with them, and what plurality or obstruction must remain explicit. This is where the architecture meets the observer classes, replay impact labels, and bounded warrants developed elsewhere in the series. High-impact revelation rights should be activated explicitly, often requiring threshold or role-separated co-authorisation.

A simple rights split makes the point concrete. The author of a private lane may hold replay and unseal rights over the full trace. A teammate may hold only the right to inspect the active certificate and summary shell. A release principal may be permitted to request wider revelation, but only under co-authorised unsealing. The architecture therefore does not ask one observer class to serve every use. It lets lawful readings differ by right, scope, and witness tier.

## 8 Open Frontier Questions

The architectural centre is stable enough that the remaining work can be stated as extension rather than as hidden prerequisite. The next frontier laws likely fall into two clusters.

**Governance and reliance laws.**

- **exposure ledger and contamination calculus**,
- collective sovereignty over co-owned traces,
- mode-transition contamination and stale certificate handling,
- **revocation propagation over dependency graphs**,
- aperture decay as a witnessed act,
- temporal-logic query languages over lawful transport, admission, and revelation paths.

**Engineering seams.**

- lawful compilation of app-authored optics into intent and observer plans,
- settlement shells and idempotent import of already-adjudicated outcomes,
- export semantics at the causal-medium boundary.

These points become visible once the foundations are coherent enough to support a real social-runtime theory. The final paper should therefore turn from the internal recurrence of the WARP optic to the external consequence of that recurrence: there is no primary runtime. If graph-like state is an observer-relative reading and replica import is ordinary witnessed admission, Continuum is best understood as the protocol-shaped causal medium defined by witnessed suffixes, transport, slicing, lowering, outcome algebra, and bounded revelation. Paper VIII develops that consequence directly by treating Continuum as a protocol of witnessed admission rather than as a privileged runtime.

## 9 Conclusion

The central result of this paper is a clearer object. WARP is an optic architecture in which state transitions, braid outcomes, certificate trust, historical folding, and observer revelation are treated as instances of the same witnessed act: project through an observer, bound an optic slice, lawfully lower a weave, and retain the result as a hologram. Because each recurrence emits such a hologram, the same downstream uses recur beyond ticks, including replay and audit, transport, and counterfactual examination. Every retained slice contains the least sufficient history of its bounded admitted situation up to the governing retention contract. Tick replay is therefore only the first concrete instance of a more general property that also appears in braid lowering, distributed import, and later trust-bearing shells.

The distributed consequence is correspondingly precise. Transport constructs the comparable slice on which replica-scale lowering can occur; once that slice exists, the downstream optic is the same one already used locally. Distribution changes the slice boundary while leaving the downstream law unchanged.

That optic still requires judgement and interpretation. It does, however, make the terms of collaboration explicit: what plural situation was lowered, over what site, under what policy, with what retained witness, and under what revelation rights. In a WARP-native system, when software executes, it commits, folds, and reveals truth under law.

# A Adversarial Transport (Sketch)

The main text treats replica transport as a normalisation step that constructs a comparable basis and re-expresses a remote suffix family as a weave suitable for lowering. This appendix records the minimal obligations we expect transport to satisfy under hostile distributed conditions. The goal is not a complete protocol or a full formalisation; it is to keep the architectural interface honest about what "transport" must supply so that the replica optic can either lower lawfully or return an explicit `Conflict` or `Obstruction`.

## A.1 Threat model

We assume:

- an untrusted network that may reorder, delay, duplicate, or drop messages;
- remote replicas that may be faulty, partially connected, or actively adversarial (including equivocation) [LSP82];
- bounded local resources, so a receiver must remain able to refuse work that exceeds its admitted slice and retention posture.

We do not assume a global reliable broadcast or a synchronous network [Bra87]. We also do not assume that all participants share the same authority regime; trust and reliance are part of the witness-bearing objects, not ambient background.

## A.2 Transport obligations

At replica scale, the lowering surface $\rho_{\text{replica}}$ uses Transport to construct a weave $\widetilde{C}$ at a receiving comparable frontier. Under hostile conditions, that construction must provide at least:

- **Authorship evidence**: a proof-carrying identity for the claimed remote suffix family and its basis anchors (for example signatures or certificate chains), sufficient for the receiver's authority regime to decide whether reliance is permitted at all.
- **Basis comparability**: an explicit description of the basis anchors and transport path used to re-express the remote claims at the receiver's comparable frontier. If comparability cannot be established, transport yields `Obstruction`.
- **Witness carriage**: a witness tier sufficient for replay, audit, and later revocation or dispute. Transport must not erase the ability to explain why a later import was admitted, blocked, or obstructed.
- **Idempotence keys**: stable identifiers for transported claim families and their derived settlement shells, so that duplicate delivery can be treated as replay of an already witnessed object.

These obligations do not solve adversarial transport. They make its inputs and outputs explicit enough that downstream lowering remains lawful and auditable.

### A.3 Failure modes and lawful outcomes

Hostile and failure-path behaviour should surface as witnessed outcomes instead of being absorbed as silent heuristics:

- **Unauthenticated claims**: transport returns `Obstruction` with a witness explaining the missing or insufficient evidence.
- **Equivocation**: when a principal authors incompatible remote suffix families over the same basis, the receiver treats the situation as a lawful `Conflict` or a policy-driven `Obstruction`, and emits a witness suitable for later revocation or quarantine.
- **Partial connectivity**: missing dependencies, missing basis anchors, or incomplete transported slices return `Obstruction` rather than speculative patch application.
- **Replay and duplication**: duplicate delivery becomes idempotent re-introduction of the same witnessed transport object, not a new authored intent.
- **Resource exhaustion**: when a transported family would force a receiver beyond its bounded slice $\chi$, retention contract $\Lambda$, or authority regime $\alpha$, the receiver may lawfully obstruct the import at the boundary.

### A.4 Architectural consequence

The replica-scale doctrine remains the same: transport constructs the comparable slice and produces a weave; lowering judges that weave under the governing optic and retention contract. Under hostile conditions, transport and authority decide whether the imported object is admissible for judgement at all. The outcome is therefore witnessed: `Derived` when import succeeds, `Conflict` when incompatibility is surfaced, and `Obstruction` when preconditions fail.

## B WARP Glossary

This glossary gathers the recurring nouns of this paper and the surrounding series. The entries are intentionally architectural and implementation-agnostic.

### B.1 Symbolic Vocabulary

| Symbol | Meaning |
|---|---|
| $\Omega = (\pi, \beta, M, U, E)$ | Observer plan: projection, basis, observer state, update discipline, and emission. |
| $\Psi = (\Omega, \chi, \rho, \Pi, \Lambda)$ | WARP optic: observer plan, optic slice, set-side lowering surface, admissibility law, and retention contract. |
| $\mathcal{P}$ | Basis-relative weave. Depending on scale, this may be a candidate rewrite bundle, a braid, or a transported import family. |
| $\rho(F_*, \mathcal{P}; \Omega) = (C, \chi)$ | Set-side lowering surface presenting comparable claims $C$ over bounded slice $\chi$ from weave $\mathcal{P}$ at frontier $F_*$. |
| $\mathrm{Lower}_\Psi(F_*, \mathcal{P}) = (R, W, \theta)$ | Full optic act: lower weave $\mathcal{P}$ under optic $\Psi$ at frontier $F_*$, yielding outcome $R$, witness $W$, and hologram $\theta$. |
| $\mathcal{O}(X) = \mathrm{Derived}(X) \sqcup \mathrm{Plural}(X) \sqcup \mathrm{Conflict} \sqcup \mathrm{Obstruction}$ | Outcome algebra for lawful lowering over scale-local carrier $X$. |
| $\theta \simeq_\Lambda \theta'$ | Shell equivalence under retention contract $\Lambda$. Equivalent shells preserve the same optic-accurate bulk for the lawful uses demanded by $\Lambda$. |
| $\mathrm{Transport}(F, \Sigma, F')$ | Replica-scale basis construction: re-express remote suffix family $\Sigma$ from frontier $F$ at comparable frontier $F'$ before lowering. |
| $\mathrm{Commit}_\Psi(F_*, \mathcal{P})$ | Commitment act: shorthand for optic lowering when the lowered result is being treated as admitted frontier-relative truth. |
| $\mathrm{Fold}_\Lambda(\theta)$ | Fold a shell into a $\Lambda$-equivalent retained carrier. |
| $\mathrm{Reveal}_{\Omega,\alpha}(m, \theta)$ | Revelation act under observer plan $\Omega$, authority regime $\alpha$, observer state $m$, and shell $\theta$. |

### B.2 History And Lane Nouns

| Term | Meaning |
|---|---|
| WARP | An optic architecture over shared causal history. The system's primary object is witnessed causal history, with frontier-relative materialisations over it. |
| Continuum | The protocol-shaped causal medium formed by exchangeable witness-bearing artefacts (suffixes, shells, receipts, certificates) and the transport rules that make remote history comparable for lawful lowering. |
| Causal lane | A history-bearing computational object. The ontology-level noun for a causal line of development, whether canonical or speculative. |
| Worldline | A canonically admitted causal lane. Its history and frontier-relative materialisations count as shared causal reality for the relevant policy frame. |
| Strand | A speculative causal lane rooted at a parent worldline and fork anchor, carrying lawful local divergence over inherited history. |
| Braid | A lane-scale weave over multiple lanes sharing a common precursor or basis, presented for comparison and later collapse under policy. |
| Weave | A basis-relative carrier that holds multiple candidate claims in suspension without yet forcing one derived answer. Candidate rewrite bundles, braids, and transported import families are the primary cases in this paper. |
| Frontier | The coordinate at which a lane or materialisation is interpreted. A frontier-relative view changes with the chosen basis. |
| Basis | The comparison or reading coordinate relative to which claims, slices, or observers are made commensurable. |
| Fork anchor | The exact inherited basis at which a strand diverges from a parent worldline. |
| Shared history | The admitted causal medium from which frontier-relative readings are materialised. |

## B.3 Admission And Replay Nouns

| Term | Meaning |
| --- | --- |
| Claim family $C$ | The comparable claim presentation produced by an optic's set-side lowering surface over a bounded slice. |
| Policy $\Pi$ | The governing law under which a bounded site is judged. |
| Optic slice $\chi$ | The least frontier-relative closure over which the optic can lawfully compare the claims and judge their coexistence, derivation, plurality, conflict, or obstruction. |
| Lowering | The set-side act by which an optic lawfully judges a weave over a bounded slice and emits outcome, witness, and hologram. |
| Derived | An outcome in which a lawful result is admitted as the derived local or imported result. |
| Plural | An outcome in which plurality itself is the lawful admitted result. |
| Conflict | An outcome in which the system surfaces genuine incompatibility as an explicit result. |
| Obstruction | An outcome in which lawful admission cannot proceed because the relevant preconditions or transport conditions are not satisfied. |
| Witness $W$ | The semantic explanation of why a lowering outcome was lawful. It behaves as proof-bearing residue that supports replay, audit, and explanation. |
| Shell $\theta$ | The retained carrier packaging a lowered result together with witness-bearing consequences for replay, audit, transport, or revelation. |
| WARP hologram | A retained shell preserving the least sufficient boundary needed to recover the optic-accurate bulk of a bounded admitted situation for the required contract. |
| Boundary Transition Record | One concrete shell family, especially at tick scale, used to retain replayable local transition boundaries. |
| Transport | The replica-scale normalisation step that re-expresses a remote suffix at a comparable frontier before import judgement. |
| Normaliser | The basis-construction step inside an optic's set-side lowering surface. At replica scale, the hard case is transport. |
| Optic-accurate bulk | The retained causal content, together with the required witness structure, that must remain recoverable for the observations and obligations named by an optic and its retention contract. |

## B.4 Commitment, Revelation, And Governance Nouns

| Term | Meaning |
|---|---|
| Optic | The composite act by which an observer bounds a site through an aperture, a weave is lawfully lowered there, and the resulting admitted situation is retained as a holographic witness in append-only history. |
| Commitment | The act by which the witnessed outcome of lawful lowering is admitted into frontier-relative truth, whether derivation, plurality, conflict, or obstruction. |
| Folding | Re-expression of already admitted history into a boundary-equivalent retained shell under an explicit contract. |
| Revelation | Lawful exposure of admitted or folded history to a bounded observer under an aperture and authority regime. |
| Observer plan $\Omega$ | The revelation-side object specifying projection, basis, accumulation state, update discipline, and emission. |
| Observer | A positioned reading discipline that determines which readings may be lawfully emitted at a given aperture, basis, and authority tier. |
| Reading | A lawful emitted chart over causal history obtained through a bounded observer frame. |
| Aperture | The bounded observational scope within which a reading is lawful. |
| Retention contract $\Lambda$ | The obligations a retained shell must continue to satisfy for replay, audit, transport, revelation, or reliance review. |
| Authority regime $\alpha$ | The active rights and reliance discipline governing revelation. |
| Shell equivalence $\simeq_\Lambda$ | Equivalence between shells relative to the retained contract $\Lambda$, defined by preservation of lawful replay, audit, transport, or revelation obligations, independent of byte identity. |
| Property certificate | A proof-bearing shell attesting that specified properties hold relative to a shared basis and witness tier. |
| Reliance status | The lifecycle state of a certificate or trust-bearing object: issued, active, superseded, suspended, revoked, expired, or archived. |
| Observer rights | The rights lattice governing who may inspect, replay, export, delegate, unseal, or revoke access over retained artefacts. |
| Ephemeral Scratch | Local, weakly retained, disposable work that has not yet crossed into durable collaborative history. |
| Author-Only Speculative Lane | Durable, replayable, but author-sealed speculative work that has not yet been socially transported. |
| Shared / Admitted Lane | The portion of history that has crossed into collaboratively admitted causal truth. |

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